

# Predicting Customer Churn when using Dynamic Pricing Models

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## 1 Introduction

Ride-sharing platforms such as Uber and Lyft have long utilized dynamic pricing. They adjust their pricing based on real-time supply and demand to maximize profitability. Now, this model is expanding to retail and grocery stores. Currently, online grocery shopping stores such as Instacart are implementing dynamic pricing into their applications to maximize profits. However, when implementing a system that increases prices, if the cost of an item exceeds the price a shopper is willing to pay, then it can trigger customer churning and drive them to competitors.

The optimization problem, although well defined, is much more difficult in practice. Each retailer would need to know for each customer at which price threshold would they churn, since some customers have a higher tolerance while others are more sensitive.

To solve this problem, we propose a solution using a machine learning model designed specifically to predict dynamic pricing churn within grocery retail. Because historical grocery retail datasets do not explicitly contain a native churn label related to personalized price hikes, we had to introduce a stochastic simulation method to model individual pricing tolerance based on historical discount. We evaluated several predictive models using Logistic Regression as our baseline against our higher performing models, Random Forest, XGBoost and XGBoost + Isotonic Calibration. Our framework identifies the threshold where the marginal gains from price increase are offset by the financial losses of customer churn, illustrating the fundamental between short-term revenue vs long-term retention.

## 2 Related Works

For similar works, in an article titled “A comparison of machine learning techniques for customer churn prediction” the author evaluated various machine learning methods to predict customer churn. They identify backpropagation and decision tree classifiers as the best methods of the five they tested using precision, recall, accuracy, and F-score as metrics. We are using similar metrics, as they seem intuitive due to the nature of the data likely having class imbalances.

Another article titled “Machine Learning-based Price Optimization for Dynamic Pricing on Online Retail” demonstrated that machine learning based dynamic pricing models can improve profitability in e-commerce. Their research shows that machine learning models are ideal for dynamic pricing. However, in their research they mainly focused on maximizing

profit, but did not account for customer retention risk. Our work addresses this gap by explicitly modeling the churn risk triggered by increased prices, which allows us to identify where the max profit along with customer retention is.

## 3 Data

We use the “Dunnhumby - The Complete Journey” dataset from Kaggle. This dataset provides household-level transactional data that represents over 2500 households. The dataset includes information that will help to understand the drivers behind each shopper. With this we will be able to simulate dynamic price fluctuations on historical data to predict which customers have the highest risk of churn when facing personalized prices.

The Dunnhumby Complete Journey dataset includes transaction-level purchases, household demographics, and coupon data. We began by removing missing and duplicate records and converting date fields into a consistent format to support time-based analysis.

Since churn is defined at the customer level, transaction data was aggregated into household-level features, including recency, purchase frequency, total and average spending, basket size, and category diversity. We also engineered features to capture price sensitivity, such as coupon redemption rate and discount usage.

Household demographic data was merged using household IDs. Missing values were imputed (median for numerical variables, mode for categorical), and categorical variables were one-hot encoded.

Because churn is not explicitly provided, we defined it as a prolonged period of inactivity (e.g., no purchases within a future time window) following simulated price increases. The final dataset was then split into training and testing sets, with class imbalance handled using appropriate techniques.

## 4 Methods

### 4.1 Churn Simulation

Since the dataset does not contain churn labels under dynamic pricing conditions, we simulate churn based on price increases.

For each customer we define price tolerance and compute churn probability using a logistic function:

$$P(churn) = 1/(1 + \exp ($$

Churn outcomes are then sampled from Bernoulli distribution.

Previously there was a more deterministic approach where all customers churn immediately once a threshold is crossed. However, this probabilistic approach makes customers more likely to churn as the price increase exceeds their tolerance, but it is not

automatic. Though it produces messier scores, it creates a more realistic scenario of how real customers behave. We can capture variability in customer responses rather than assuming deterministic behavior. Some may stay longer than expected due to habits, loyalty, etc.

## 4.2 Train Test Split

All models are implemented using a preprocessing pipeline to ensure consistency in feature transformation.

Churn problems usually have many non-churners and few churners. This creates a class imbalance that pushes the models to predict no churn for everyone. To combat this, we apply class weighting during training.

The dataset is split temporally into training and testing sets so that later weeks are treated as unseen future behavior.

## 4.3 Logistic Regression

Logistic regression is used as a baseline model. It assumes a linear relationship between features and the log-odds of churn. This model provides interpretability which allows us to identify which factors are most strongly associated with customer churn.

## 4.4 Random Forest

Random Forest is an ensemble model that constructs multiple decision trees and averages their predictions. It is used to capture nonlinear relationships and interactions between features. This is particularly useful in this setting since customer responses to pricing changes may show threshold effects.

In hyperparameter tuning, choices were made based off common practice and what best fits our scenario. The number of estimators was set to 200 in order to ensure stability without excessive training time. A max depth of 8 was set to prevent overfitting while still capturing nonlinear relationships between pricing changes and customer behavior. We fix a random seed to ensure reproducibility of results.

## 4.5 XGBoost

Gradient boosting builds trees sequentially to improve prediction of accuracy. Each tree corrects errors from previous models. This model is expected to provide stronger predictions than the random forest method.

## 4.6 XGBoost + Isotonic Calibration

We built this model using the base XGBoost model, though models like these often output uncalibrated probabilities that make the default 0.5 decision threshold unreliable. To correct this, we further split our training data by using 80% to train the base XGBoost model and dedicate the remaining 20% for calibration. We then applied Isotonic Calibration on the calibration set to adjust the model's scores into more reliable probability estimates that would better reflect the true churn rates. Finally, we evaluated the precision-recall curve on

the calibration set to find the optimal threshold that maximized the F-1 score to replace the arbitrary 0.5 default for our final predictions.

## 5 Experiments and Results

### 5.1 Model Performance

Performance is evaluated using accuracy, precision, recall, F1-score, AUC, and Brier. These are demonstrated in both tabular and graphical form.

| Model                           | Accuracy | Precision | Recall | F1     | AUC    | Brier  |
|---------------------------------|----------|-----------|--------|--------|--------|--------|
| Logistic Regression             | 0.6441   | 0.4966    | 0.4877 | 0.4921 | 0.6425 | 0.2247 |
| Random Forest                   | 0.7763   | 0.6517    | 0.7886 | 0.7137 | 0.8522 | 0.1585 |
| XGBoost                         | 0.7673   | 0.6290    | 0.8327 | 0.7167 | 0.8596 | 0.1468 |
| XGBoost + Isotonic ( $t=0.37$ ) | 0.7673   | 0.6290    | 0.8327 | 0.7167 | 0.8596 | 0.1396 |

Model Comparison — Predicting Customer Churn Under Dynamic Pricing

Figure 1: Table of Metrics

Tree-based models outperform logistic regression. This suggests that customer churn behavior under dynamic pricing is influenced by nonlinear relationships.

### 5.2 Price Increases and Churn

We analyze how churn probabilities change under different simulated price increases. For each price level the average predicted churn probabilities are computed across customers.

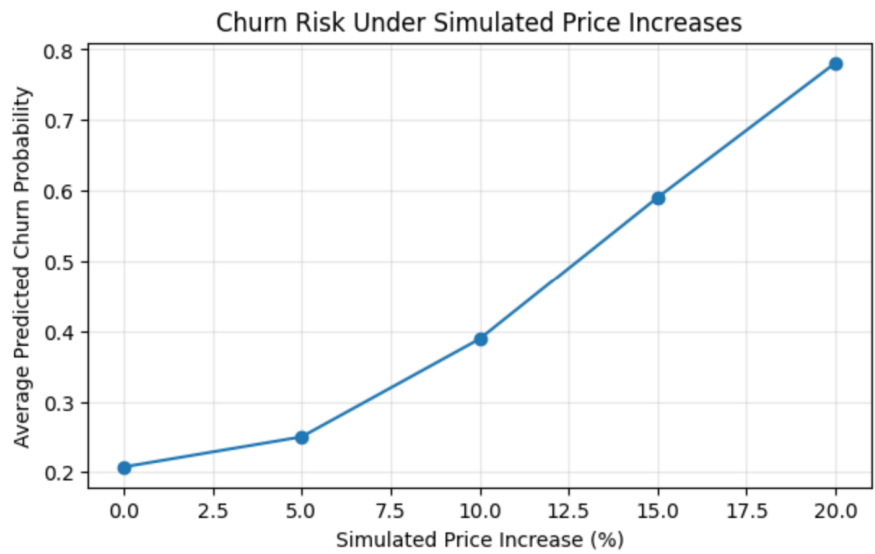


Figure 2: Price Increase vs Predicted Churn

As shown in Figure 2, churn probability increases as price increases. This relationship, however, is nonlinear. This is consistent with the expected behavior of where larger price increases lead to higher churn

### 5.3 Revenue Trade-off

As mentioned in the introduction, we are interested in evaluating the impact of pricing on business outcomes. We do this by estimating expected revenue under different price scenarios.

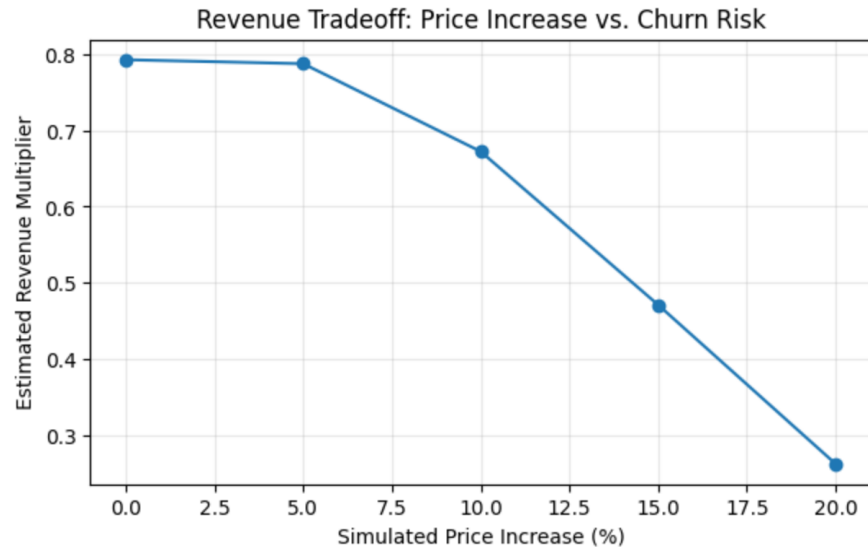


Figure 3: Price Increase vs Revenue Multiplier

We see that price small increases keep stable revenue at first, but excessive price increases reduce overall revenue as it leads to higher churn. This shows the trade-off between maximizing revenue and maintaining customers.

## 6 Conclusion

In this project, we simulate customer churn under dynamic pricing and evaluate machine learning models for predicting churn behavior. Our results show that nonlinear models outperform the linear model, indicating that customer responses to price changes are complex.

We also find that while price increases can improve revenue, excessive increases lead to significant churn. These findings demonstrate the importance of balancing pricing strategies with customer retention.

Future work could explore more advanced behavioral modeling and incorporate additional external factors such as competitor pricing and current events.

## 7 References

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